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ANALYSIS IN GEOCHEMISTRY
FOR ELEMENTS

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CURRENT GEOELECTRIC SOUNDING

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X-RAY SPECTROMETRY OF ROCKS

W. ERNST
CHEMICAL FACIES ANALYSIS

P.V. SHARMA
PHYSICAL METHODS IN GEOLOGY

C.H. CHEN (Editor)
COMPUTER-AIDED SEISMIC ANALYSIS AND DISCRIMINATION

D. KOEFOED
SOUNDING PRINCIPLES, 1. RESISTIVITY SOUNDING
EXPERIMENTS

H.P. PATRA AND K. MALLICK
SOUNDING PRINCIPLES, 2. TIME-VARYING GEOELECTRIC
SOUNDINGS

Methods in Geochemistry and Geophysics

3

MINING GEOPHYSICS

BY

D.S. PARASNIS

*Professor of Applied Geophysics
University of Lulea, Lulea, Sweden*

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TABLE I
SUSCEPTIBILITIES OF SAMPLES OF ROCKS AND ORES
(Rationalised units)

Rocks	Susceptibility $\times 10^6$	Locality	Observer	Remarks
Limestone	500–750	Berggieshübel, Germany	Neumann (1932)	some Fe_3O_4
Limestone	12.5–24,700	Various, Sweden	Werner (1945)	
Diabase tuffs	50,000–245,000	South Bahrtal, Germany	Neumann (1932)	metamorphic
Diabase tuffs	1,900–2,380	North Bahrtal, Germany	Neumann (1932)	metamorphic
Diabase	10,000–150,000		Mooney and Bleifuss (1953)	0–3.4% magnetite 0–0.9% ilmenite
Basalt	250–105,000		Mooney and Bleifuss (1953)	0–2.5% magnetite 0–0.6% ilmenite
Rhyolite	250–9,500		Mooney and Bleifuss (1953)	0–0.4% magnetite
Gabbro	1,000–76,000		Mooney and Bleifuss (1953)	0–0.9% magnetite 0–0.2% ilmenite
Granite	0–49,500		Mooney and Bleifuss (1953)	0–1.3% magnetite
Greenstone	500–11,000		Mooney and Bleifuss (1953)	0–0.2% magnetite
Slates	0–1,250		Mooney and Bleifuss (1953)	0–0.2% magnetite
Skarn	500–5,300	Ställberg, Sweden	Werner (1945)	
Granulite	88–46,300	Persberg, Sweden	Werner (1945)	
Porphyry	225–210,000	Kiruna, Sweden	Werner (1945)	

TABLE I (continued)

Ores	Susceptibility $\times 10^6$	Locality	Observer	Remarks
Brown iron ore	1,000–1,750	Berggieshübel, Germany	Neumann (1932)	haematite
Haematite	6,630	Långban, Sweden	Werner (1945)	rich, very low magn.
Haematite	4,110–9,860	Striberg, Sweden	Werner (1945)	low magn.
Magnetite ore	71,000–97,000	Berggieshübel, Germany	Neumann (1932)	
Magnetite ore	3.0×10^6 – 6.15×10^6	Ställberg, Sweden	Werner (1945)	31.1–63.3% magnetite
Magnetite ore	1.25×10^7 – 1.4×10^7	Grängesberg, Sweden	Werner (1945)	86.1–94.6% magnetite
Chromite ore	7,500–1.2 $\times 10^6$	Various	Hawkes (1951)	26.76–58.40% Fe
Iron ore	6.5×10^6 – 1.53×10^6	Biwabik and Soudan, U.S.A.	Mooney and Bleifuss (1953)	15.4–26.1% magnetite
Hausmannite	1,650	Långban, Sweden	Werner (1945)	Mn_3O_4
Jacobsite	25,000	India	Bhimsankaram and Rao (1958)	(MnFe_3O_4)
Pyrite ore	5,300	Kankberg, Sweden	Malmqvist (1960)	compact
Pyrrhotite	750	Kankberg, Sweden	Malmqvist (1960)	compact

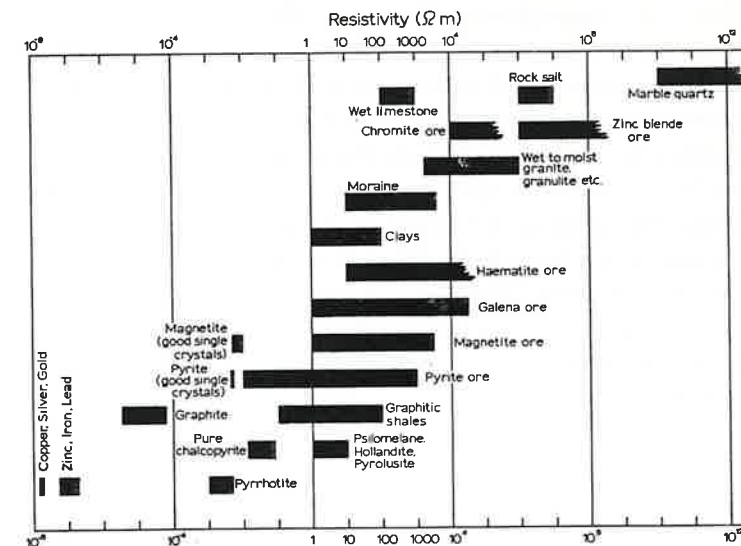


Fig.56. Approximate resistivity ranges of some rocks, minerals, ores and common metals. The resistivity scale (horizontal) is logarithmic.

of the order of 20,000–50,000 Ωm (in situ) but moderate impregnation of conducting minerals brings its resistivity down to 3,000–5,000 Ωm in many areas.

Fig.56 shows the approximate resistivity ranges of various ores and rocks. A fuller discussion of the resistivity of ores will be found in Parasnis (1956).

APPARATUS

The apparatus required for electrical prospecting need not be complicated and is therefore easily rigged up, unless a number of refinements like automatic controls are desired. In fact, very satisfactory electric surveys can often be carried out with the simple equipment consisting of high-tension dry batteries as the source of electric current, four metal stakes – two as current electrodes and two as voltage probes – a milliammeter, a voltmeter and a sufficient length of well insulated cable.

Suitable batteries for the work are the so-called anode batteries giving 45 or 90 V per battery. Two or three such batteries are connected in series so as to give a total voltage of some 135–180 V. Also included in the circuit are a mA-meter, a switch and an adjustable resistance to regulate the current (Fig.57), the whole assembly being neatly mounted in a small box.

Anode batteries do not have a very great current capacity and they need frequent replacement. Hence, for large scale surveys it is found more convenient to use a petrol-driven direct-current generator supplying, say, 200 W–1 kW power. Such a generator cannot, of course, be carried along like a battery box but must be kept stationary and connected to the current electrodes by long cables.

The mA-meter should be capable of reading, say, 100 mA full scale but suitable shunts should be provided so that its measuring range can be extended if necessary. Usually, manufacturers will supply the necessary shunts for a mA- or a μA -meter. Universal type instruments such as the AVO-meter, Simpson, Philips, Unigor etc., which have several current ranges, can also be employed for the measurement of current.

The current electrodes may be round stainless-steel rods, about 70 cm long, with a provision (like a clip) for fastening the uncovered end of the cable leading to the battery. A satisfactory electrical contact will usually be obtained on reasonably moist ground if the rods are

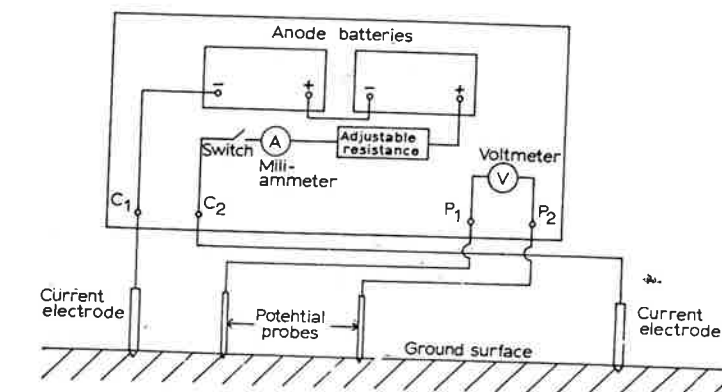


Fig.57. Scheme of simple field apparatus for resistivity measurements.

The density corresponding to that curve which shows the least correlation with topography is adopted as the density of the slab.

In the straight-line method the approach is somewhat different and more objective. Suppose that the Bouguer anomaly at the stations along a line is exactly zero. Then eq. (8.6) can be written, with a little rearrangement of terms, as:

$$\Delta g_{obs.} \pm \Delta g_{\Phi} = + (0.4191 h - T)\rho - 3.086 h \tag{8.7}$$

If we denote the left-hand side by y and the term $+(0.4191 h - T)$ by x we see that the equation is of the form:

$$y = \rho x + c$$

which is the equation of a straight line whose slope is ρ . Thus, if we plot the measured quantity $\Delta g_{obs.} \pm \Delta g_{\Phi}$ along a profile against x , and draw a straight line through the points, the slope of this line will yield the density of the slab. In practice, the points will not all lie exactly on a straight line because the Bouguer anomaly will not in general be zero. The statistical expedient of least squares can be used in this case for drawing the best straight line.

The subject of field determinations of density is more complicated than the above description would indicate and the interested reader should refer to the extensive literature on this subject (Nettleton, 1939; Hammer, 1950; Cook and Thirlaway, 1952; Parasnis, 1952; Domzalski, 1955; Vajk, 1956; Whetton et al., 1957; Jung, 1959; Grant and Elsharty, 1962).

Table V summarizes the density data on some of the important rocks and ores, but the values indicated should not, of course, be looked upon as substitutes for rock density measurements within the survey area.

In the *Système Internationale* (SI) the unit of density is kg/m^3 which is a thousand times larger than the unit g/cm^3 . It is recommended in the SI that additional units may be created by multiplication of any basic units by powers of ten. If then we multiply the SI unit by 1000 we get the unit megagramme/ m^3 (=tonne/ m^3) and this

TABLE V

DENSITIES¹ OF ROCKS AND ORE MINERALS IN PURE FORM

	Density (g/cm ³)
Sand, dry	1.40–1.65
Sand, wet	1.95–2.05
Coal	1.20–1.50
English chalk	1.94
Rock salt	2.10–2.40
Serpentine	2.50–2.55
Limestone	2.60–2.70
Quartzite	2.60–2.70
Gneiss	2.70
Granite	2.50–2.70
Anhydrite	2.96
Diabase	2.50–3.20
Basalt	2.70–3.30
Gabbro	2.70–3.50
Zinc blende	4.0
Chalcopyrite	4.2
Chromite	4.5–4.8
Pyrrhotite	4.6
Pyrite	5.0
Haematite	5.1
Magnetite	5.2
Galena	7.5

¹ The density of an ore will depend upon the relative amounts of gangue and ore minerals.

is exactly the same as g/cm^3 . Where rock and ore masses are concerned it is far more appropriate and realistic to talk of the density as Mg/m^3 than as g/cm^3 . On account of the convenient size of the numbers obtained when density is expressed as Mg/m^3 (g/cm^3) all the formulas appearing in this book in connection with gravity work imply that the density is expressed as Mg/m^3 .